



HYEYOON JEONG

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Research Part
Energy Efficiency
Enhancement in RIS-
Assisted Systems
AI-Based Wireless
Communication Link
Optimization

Education

Sejong University Graduate School M.S. in Information &
Communication Engineering, GPA : 4.5/4.5 2025.03 ~ 2027.02
Sejong University B.S. in Electronic Information & Communication
Engineering, Major GPA : 4.13/4.5 2021.03 ~ 2025.02
Exchange Student, Kaunas University of Technology, Lithuania
2024.01 ~ 2024.06
Masan Girl's High School 2016.03 ~ 2019.02

Language Certification

OPIc IM2 | 2026.05

Research Performances

I. SCI Publication – First Author

- I. A Two-Stage Framework for Energy-Efficient Group
Activation in RIS-Assisted Communication
Systems, 2026.07

II. SCI Publication – Co-Author

- I. A Profile-Aware Resource Allocation Framework for RAN
Slicing in Cell-Free Massive MIMO Networks, 2026.07
- II. Improved Connected-Mode Discontinuous Reception
(C-DRX) Power Saving and Delay Reduction Using
Ensemble-Based Traffic Prediction, 2025.03

II. Domestic Patent

- I. Apparatus and Method for Adaptive Group Control in a
Reconfigurable Intelligent Surface-Based MU-MISO
System, 2026.04
- II. System and Method for Optimizing Transmission
Efficiency and Bit Error Rate(BER) Using Reconfigurable
Intelligent Surface(RIS) in a Multi-Base Station
Environment, 2025. 07

Conferences

I. International Conferences

- I. Energy-Efficient RIS Group Control via Random Forest with EE-Gating in Multi-User MISO Systems, 2026 ICCEE, 2026.06
- II. Learning-Based On-Off Scheduling of Reconfigurable Intelligent Surface Element Groups in MU-MISO Improving Energy Efficiency, 2026 EESP, 2026.01
- III. Enhancement of C-DRX Efficiency with Traffic Prediction Using Ensemble Learning Models, 2025 ICUFN, 2025.07

II. Domestic Conferences

- I. Energy Efficiency Analysis of Interference-Mitigation-Based Group On-Off RIS for MU-MISO Downlink, The 2026 KICS Winter Conference, 2026.02
- II. A Study on the Traffic-Aware RIS On-Off Control for Energy Efficiency, The 2026 KICS Fall Conference, 2025.11

Projects

I. Development of dynamic trust connection and intelligent management technology for hybrid multi-layered network [2nd year]

Industry-university project | Ministry of National Defense | 2026.03 - 2026.12

I. Project Topic

Development of Multimodal Information Refinement and Preprocessing Technology for Network Anomaly Cause Classification Models

II. Project Details

Anomaly Detection and Fault Type Analysis Based on Composite State Information in PN-NPN Interworking 5G Core Environment

II. Graduate School of Virtual Convergence (Sejong University) [3rd year]

Industry-university project | IITP | 2025.01 - 2025.12

I. Project Topic

Research on Metaverse-Converged Intelligent Mobility Technology

II. Project Details

Follow-up Research Linked to Industry-Academia Cooperation Project: RIS Wireless Link Reliability and Performance Optimization for Real-Time Synchronization and Control Data Transmission Support